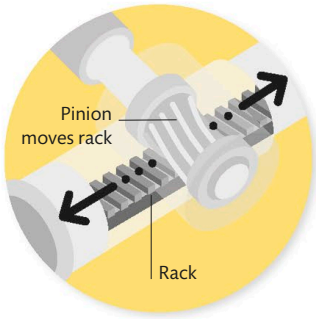
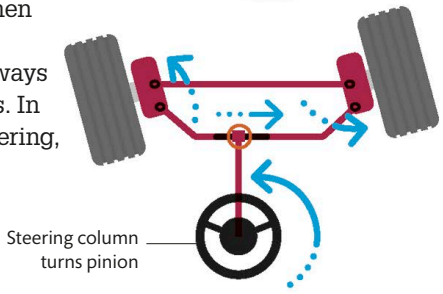
Steering

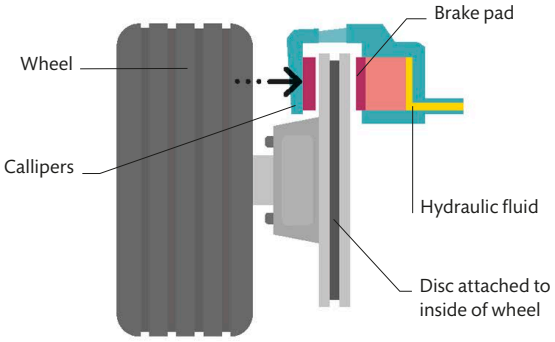
The simplest steering systems in cars rely on a type of gear mechanism called a rack and pinion. Turning a car's steering wheel rotates a pinion—a small, round gear. Its teeth engage the teeth on a flat bar called a rack. When the pinion turns, it moves the rack sideways and turns the wheels. In a car with power steering, high-pressure oil or electric motors help to move the rack.

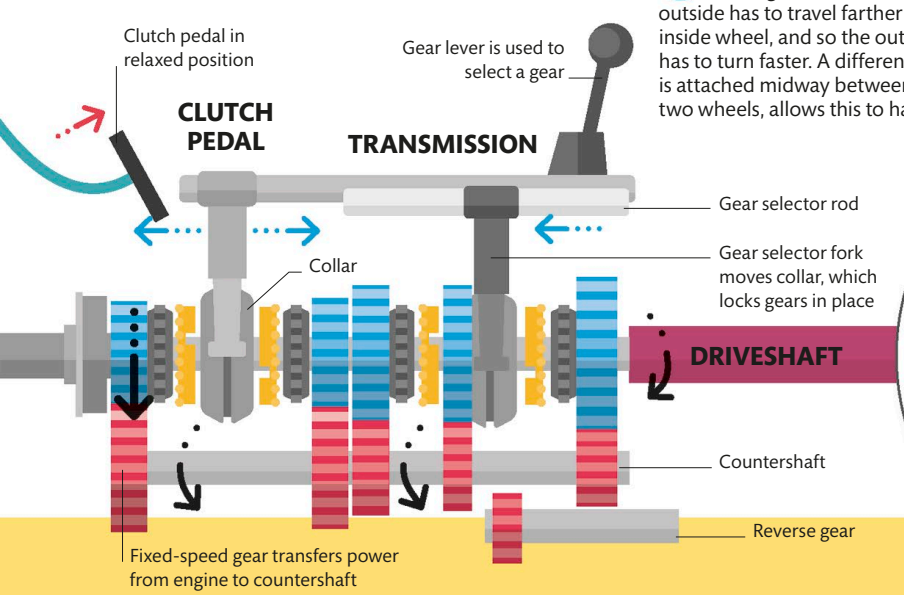




Braking

Most cars have disc brakes. A disc is fixed to each wheel, and when the wheel spins, so does the disc. When the driver presses the brake pedal, hydraulic fluid forces brake pads, mounted on callipers, to push against the disc to slow the wheel down.





3 Transmission

Piston engines work most efficiently at relatively high speeds, so gears are necessary to reduce this to the slower rotational speed needed for the car's wheels. Each gear gives a different range of speeds. First gear is normally selected to get a car moving.

4 Driveshaft

In a rear-wheel-drive car, a long driveshaft connects the transmission to the rear wheels. In a front-wheel-drive car, the engine sits above the front wheels and turns them via a short driveshaft, differential, and half-shafts.

5 Differential

During a turn, the wheel on the outside has to travel farther than the inside wheel, and so the outer wheel has to turn faster. A differential, which is attached midway between the two wheels, allows this to happen.

